

Moein Razavi, Ph.D.

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[Google Scholar](#) | [GitHub](#) | [LinkedIn](#) | [Portfolio](#)

RESEARCH INTERESTS

AI-driven disaster and urban resilience · Multi-modal machine learning for crisis sensing and response · Graph neural networks and foundation models for infrastructure networks · Spatial machine learning for hazard mapping and community vulnerability assessment · Reasoning models and agentic AI for disaster decision support · Retrieval-augmented generation (RAG) over tabular and geospatial data · Vision-language models for post-disaster damage assessment · Real-time physiological and environmental monitoring with wearable/IoT systems · Deep learning for time-series forecasting in energy and critical infrastructure

EDUCATION

Ph.D., Industrial Engineering (Data Science Concentration)

Aug 2021 – Dec 2023

[Texas A&M University, College Station, TX](#)

Dissertation: “Advancing Stress Detection: Real-Time, Naturalistic, and Personalized AI Approaches for Stress and Mental Disorder Detection”

GPA: 4.0/4.0 · Dean’s List · Graduate Research Fellowship · **Advisor:** Dr. Farzan Sasangohar

Coursework: Machine Learning, Statistical Modeling, Data Mining, Optimization, Healthcare Analytics

Aug 2019 – May 2021

M.S., Computer Science

[Texas A&M University, College Station, TX](#)

GPA: 4.0/4.0 · Dean’s List · Graduate Research Fellowship

Coursework: Deep Learning, Computer Vision, NLP, Reinforcement Learning, Algorithm Design, Pattern Recognition, Database Systems, Software Engineering, Data Mining

RESEARCH EXPERIENCE

Research Assistant — Applied Cognitive Ergonomics Lab

Sep 2021 – May 2024

[Texas A&M University, College Station, TX](#) | **Advisor:** Dr. Farzan Sasangohar

- Designed a **dual-branch transformer architecture** with 1D-CNN tokenizer, rotary positional encoding (RoPE), cross-attention fusion, stochastic depth, and focal loss for wearable stress detection from Apple Watch heart rate; achieved **AUC 0.993** and **F1 0.965** in 5-fold group cross-validation with on-device deployment via Core ML
- Engineered **18-feature HRV extraction pipeline** (time-domain statistics, frequency-domain LF/HF power) from photoplethysmography signals, unifying 5 heterogeneous study datasets with Mixup/CutMix augmentation and asymmetric focal loss
- Published **comprehensive scoping review** of 200+ studies on ML/DL techniques for stress detection (**79 citations**), synthesizing best practices for data preprocessing, feature engineering, and model selection in mental health monitoring
- Deployed **computer vision system** for COVID-19 safety compliance—physical distancing and mask detection—using deep learning object detection, achieving **98% accuracy** monitoring **50K+ daily interactions** on campus (**76 citations**)
- Co-authored **spatial ML study** combining unsupervised and supervised learning to identify county-level COVID-19 risk factors across the US, integrating demographic, socioeconomic, and spatial features (**9 citations**, *SERR*)
- Developed iOS app (*GlucoseCoach*) with Flask backend for real-time personalized glucose monitoring; applied **explainable AI** (SHAP, LIME) to clinical ML models for interpretable healthcare decision-making

Research Assistant — Dept. of Psychological & Brain Sciences

Texas A&M University, College Station, TX | Advisor: Dr. Takashi Yamauchi

- Created **OpenSync**—open-source platform for real-time synchronization and analysis of **multi-modal neuroscience data** (EEG/ERP, eye-tracking, biosensors) via WebSocket pipelines; adopted by **3 research labs (31 citations)**, *J. Neuroscience Methods*)
- Implemented advanced **signal processing algorithms** for EEG data preprocessing, filtering, and feature extraction; integrated eye-tracking and EEG for VR-based spatial ability assessments (**5 citations**, *IHSI 2024*)
- Built **Conv-LSTM deep learning model** for gait analysis achieving 93% accuracy in anomaly detection and fall risk prediction from time-series sensor data
- Developed **ML-based mouse-tracking system** for cognitive conflict detection and ADHD diagnosis, published at ACII 2019 (**19 citations**); co-authored studies on emotion assessment via cursor tracking (**19 citations**)
- Contributed to **VR-based behavioral intervention** for reducing risk habituation to struck-by hazards in construction, published in *iScience* (**20 citations**) and *Journal of Vision*

INDUSTRY EXPERIENCE

Aug 2023 – Present

Senior Data Scientist & AI/ML Engineer

Ford Motor Company, Detroit, MI

- Architected **multi-agent AI system** with LangChain, **RAG over tabular data** (BigQuery), chain-of-thought reasoning, and vector databases (FAISS, ChromaDB)—SQL Agent, Document Agent, LLM Decider—serving EV analytics across **10 US states**
- Developed **Generative AI reasoning agent** using LLMs for automated code complexity analysis with GCP cost estimation, reducing manual review time by 60%
- Trained **transformer-based time-series forecasting** models (TFT, Autoformer, Informer) with Hugging Face and Optuna hyperparameter optimization; improved EV predictions by 10% and cut training time by 40% via distributed GPU computing on Vertex AI
- Built **XGBoost regression models** for EV energy bill prediction by engineering features from vehicle telematics, utility rates, parking behavior, and home energy consumption (**$R^2 = 0.85$**)
- Engineered **large-scale data pipelines** processing 10+ TB of EV telemetry using PySpark, BigQuery, and Airflow DAGs; built Looker Studio dashboards automating KPI reporting for 500+ stakeholders
- Deployed production **MLOps infrastructure** on GCP (Cloud Run, Vertex AI, Dataproc) with CI/CD (Tekton, Airflow), Terraform IaC for multi-environment deployments, achieving 99.9% uptime

Jun 2023 – Sep 2024

Software Engineer & AI/ML Researcher

CereVu Medical, Inc., San Francisco Bay Area, CA

- Designed **2 SwiftUI iOS applications** for real-time biosignal monitoring with CoreUSB/CoreBluetooth hardware integration, embedded **ROPA pain detection algorithm**, and charting of 9 biomarkers
- Deployed **deep learning models** (LSTM, CNN) on wearable medical patches, improving patient monitoring accuracy by 15% across 400+ patients in 5 clinical studies
- Built scalable Python microservices (Flask, FastAPI) with WebSocket for biosignal streaming handling **10K+ concurrent connections**; created React/Next.js clinical analytics dashboard
- Designed **MLOps CI/CD pipelines** (Tekton, AWS CodeDeploy) with Docker, achieving 30% faster release cycles with zero-downtime deployments and automated rollbacks

PUBLICATIONS

Journal Articles & Book Chapters

9. **Razavi, M.**, Ziyadidegan, S., Mahmoudzadeh, A., Kazeminasab, S., et al. "Machine Learning, Deep Learning, and Data Preprocessing Techniques for Detecting, Predicting, and Monitoring Stress and Stress-Related Mental Disorders: Scoping Review." *JMIR Mental Health*, 11(1), e53714, 2024. **[79 citations]**
8. Salehi, F., **Razavi, M.**, Smith, M., Dixit, M. "Integrated Eye-Tracking and EEG Data Collection and Synchronization for VR-Based Spatial Ability Assessments." *Intelligent Human Systems Integration (IHSI)*, Vol. 119, pp. 348–360, 2024. **[5 citations]**
7. Kim, N., Grégoire, L., **Razavi, M.**, Yan, N., Ahn, C.R., Anderson, B.A. "Virtual Accident Curb Risk Habituation in Workers by Restoring Sensory Responses to Real-World Warning." *iScience*, 26(1), 2023. **[20 citations]**
6. Janfaza, V., Weston, K., **Razavi, M.**, Mandal, S., et al. "Mercury: Accelerating DNN Training by Exploiting Input Similarity." *IEEE Int'l Symposium on High-Performance Computer Architecture (HPCA)*, 2023. **[11 citations]**
5. **Razavi, M.**, McDonald, A., Mehta, R., Sasangohar, F. "Evaluating Mental Stress Among College Students Using Heart Rate and Hand Acceleration Data from Wearable Sensors." *arXiv:2309.11097*, 2023. **[26 citations]**
4. Ziyadidegan, S., **Razavi, M.**, Pesarakli, H., Javid, A.H., Erraguntla, M. "Factors Affecting the COVID-19 Risk in the US Counties: An Innovative Approach by Combining Unsupervised and Supervised Learning." *Stochastic Environmental Research and Risk Assessment*, 36(5), 1469–1484, 2022. **[9 citations]**
3. **Razavi, M.**, Janfaza, V., Yamauchi, T., et al. "OpenSync: An Open-Source Platform for Synchronizing Multiple Measures in Neuroscience Experiments." *Journal of Neuroscience Methods*, 369, 109458, 2022. **[31 citations]**
2. **Razavi, M.**, Alikhani, H., Janfaza, V., Sadeghi, B., Alikhani, E. "An Automatic System to Monitor the Physical Distance and Face Mask Wearing of Construction Workers in COVID-19 Pandemic." *SN Computer Science*, 2022. **[76 citations]**
1. Kazeminasab, S., Sadeghi, N., Janfaza, V., **Razavi, M.**, et al. "Localization, Mapping, Navigation, and Inspection Methods in In-Pipe Robots: A Review." *IEEE Access*, 9, 162035–162058, 2021. **[85 citations]**

Peer-Reviewed Conference Papers

7. Grégoire, L., Kim, N., **Razavi, M.**, Yan, N., Ahn, C., Anderson, B. "Reducing Risk Habituation to Struck-By Hazards in a Road Construction Environment Using Virtual Reality Behavioral Intervention." *Journal of Vision*, 22(14), 4180, 2022.
6. **Razavi, M.**, Ziyadidegan, S., Sasangohar, F. "Machine Learning Techniques for Prediction of Stress-Related Mental Disorders: A Scoping Review." *Proc. HFES Annual Meeting*, 66(1), 2022. **[14 citations]**
5. Ziyadidegan, S., **Razavi, M.**, Sasangohar, F. "Analyzing Physiological Responses to Quantify Anxiety Disorders: A Scoping Review." *Proc. HFES Annual Meeting*, 66(1), 2022. **[7 citations]**
4. Kazeminasab, S., Janfaza, V., **Razavi, M.**, Banks, M.K. "Smart Navigation for an In-Pipe Robot Through Multi-Phase Motion Control and Particle Filtering." *IEEE Int'l Conf. Electro-Information Technology*, 2021. **[22 citations]**
3. Leontyev, A., Yamauchi, T., **Razavi, M.** "ML-SST: ML-Based Mouse Tracking Enhances Adult ADHD Diagnosis." *8th Int'l Conf. Affective Computing and Intelligent Interaction (ACII)*, 2019. **[19 citations]**
2. Yamauchi, T., Leontyev, A., **Razavi, M.** "Assessing Emotion by Mouse-Cursor Tracking: Theoretical and Empirical Rationales." *8th Int'l Conf. ACII*, 2019. **[19 citations]**
1. Yamauchi, T., Leontyev, A., **Razavi, M.** "Mouse Tracking Measures Reveal Cognitive Conflicts Better than Response Time and Accuracy Measures." *Proc. Cognitive Science Society*, 41, 2019. **[16 citations]**

TECHNICAL SKILLS

ML & Deep Learning	PyTorch, TensorFlow, scikit-learn, XGBoost, Hugging Face Transformers, time-series forecasting (TFT, Autoformer, Informer), CNNs, LSTMs, Conv-LSTMs, attention mechanisms, GNNs, Optuna, SHAP/LIME
Generative AI & LLMs	GPT-4, Claude, Gemini, LangChain, LangGraph, CrewAI, AutoGen, OpenAI Agents SDK, RAG (FAISS, ChromaDB, Pinecone), Chain-of-Thought, prompt engineering, fine-tuning
Computer Vision & NLP	OpenCV, object detection (YOLO, Faster R-CNN), image segmentation, Hugging Face NLP, text classification, sentiment analysis, vision-language pipelines
Programming	Python (Expert), Swift/SwiftUI, TypeScript/JavaScript, Java, SQL, C++, R, MATLAB
Data & Big Data	BigQuery, PySpark, Apache Airflow, Hadoop, ETL/ELT pipelines, pandas, NumPy, SciPy, data warehousing, query optimization
Cloud & MLOps	GCP (Vertex AI, Cloud Run, Dataproc), AWS (S3, EC2, SageMaker), Azure (OpenAI, Cognitive Services), Docker, Kubernetes, Terraform, CI/CD (Tekton, GitHub Actions)
Web & App Development	FastAPI, Flask, Django, React, Next.js, Spring Boot, SwiftUI, Core ML, REST, WebSocket, GraphQL, gRPC
Signal Processing	EEG/ERP analysis, HRV feature extraction, biosignal filtering, multi-sensor fusion, photoplethysmography (PPG), wearable data pipelines

AWARDS & HONORS

- **Graduate Research Fellowship**, Texas A&M University, 2019–2023
- **Dean's List** (all semesters), Texas A&M University, 2019–2023
- **Perfect GPA (4.0/4.0)** across both Ph.D. and M.S. programs, Texas A&M University

CERTIFICATIONS

Deep Learning Specialization – DeepLearning.AI · **Machine Learning Specialization** – Stanford · **NLP Specialization** – DeepLearning.AI · **Advanced MLOps** – Coursera · **Generative AI with LLMs** – AWS · **Bayesian Statistics** – Coursera

PROFESSIONAL SERVICE & MEMBERSHIPS

- Peer reviewer for research publications in machine learning and healthcare analytics
- Member, Institute of Electrical and Electronics Engineers (IEEE)
- Member, Human Factors and Ergonomics Society (HFES)